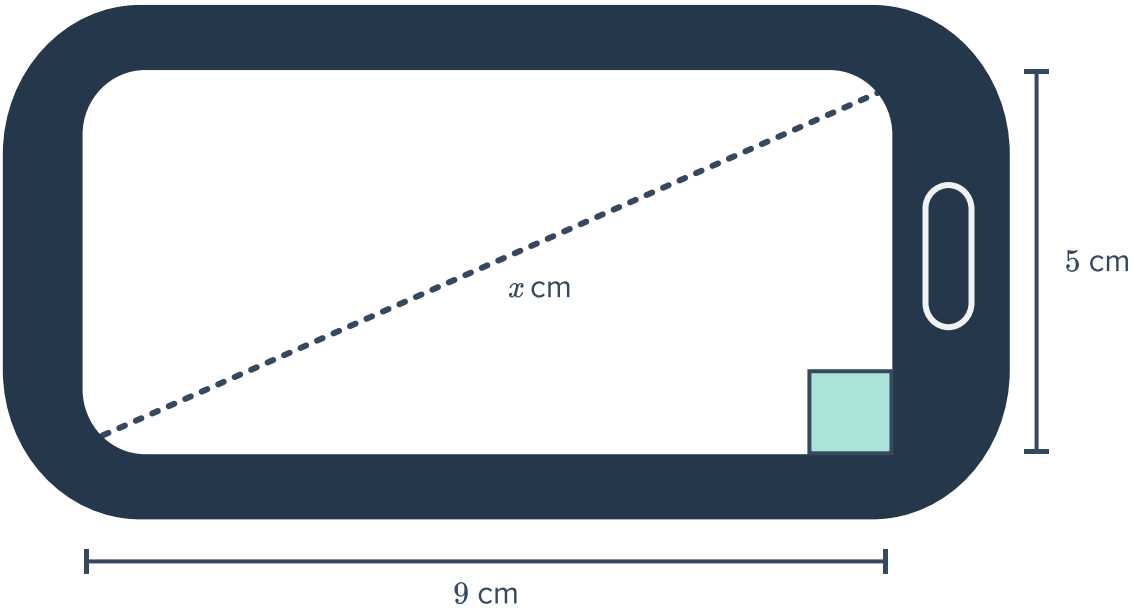
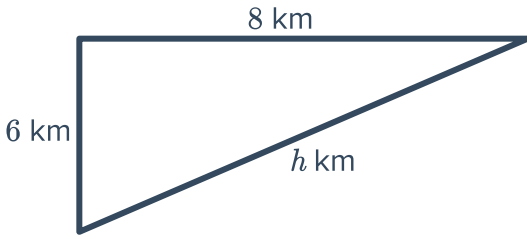
Applications of Pythagoras' theorem

1 The screen on a handheld device has dimensions $9\ \text{cm}$ by $5\ \text{cm}$, and a diagonal of length $x\ \text{cm}$. Find the value of x, correct to two decimal places.



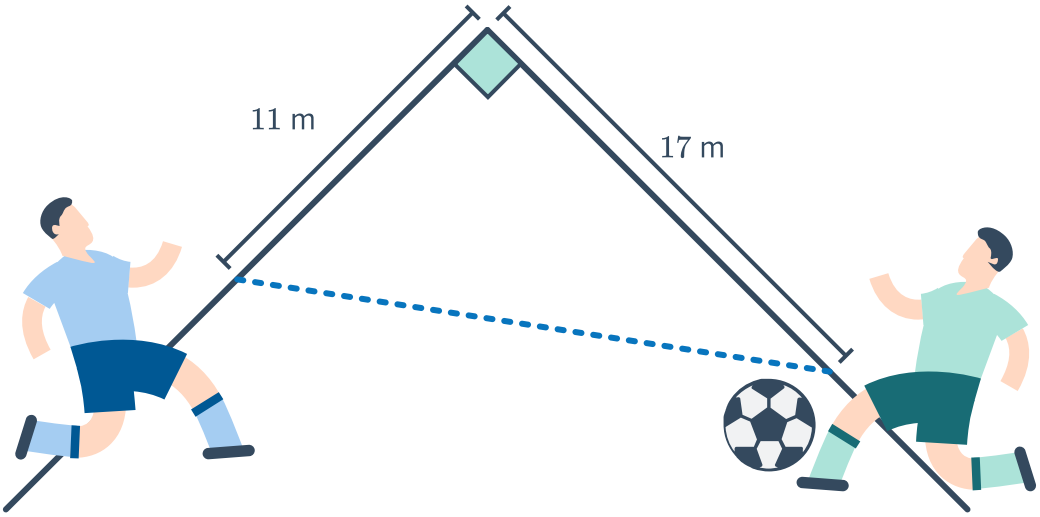
2 Glen's car has run out of petrol. He walks $8\ \text{km}$ west and then $6\ \text{km}$ south looking for a petrol station.

If he is now $h\ \text{km}$ directly from his starting point, find the value of h.



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3 Fred and Carl are playing football together. At one point in the game, they are near the same corner of the field. Fred is on the goal line, $11\ \text{m}$ away from the corner, while Carl is on the side line, $17\ \text{m}$ away from the corner.

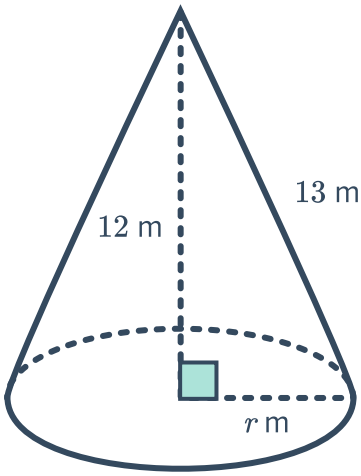


Find the shortest distance between Fred and Carl, correct to two decimal places.

4 Consider a cone with slant height $13\ \text{m}$ and perpendicular height $12\ \text{m}$:

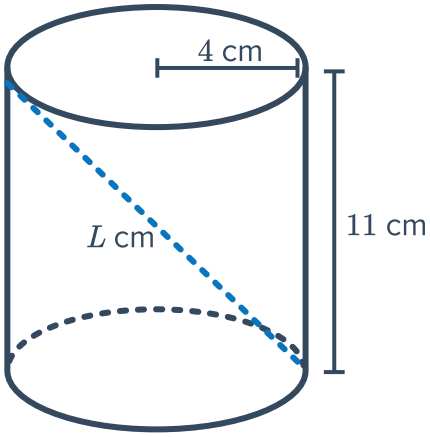
a Find the length of the radius, r, of the base of this cone.

b Hence, find the diameter of the base of the cone.



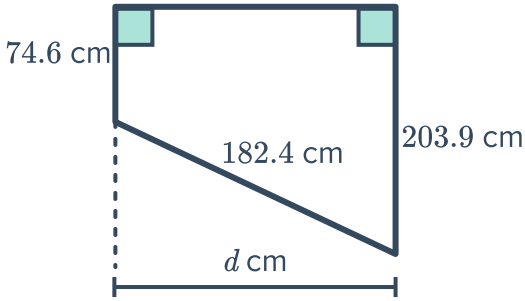
5 A soft drink can has a height of $11\ \text{cm}$ and a radius of $4\ \text{cm}$.

Find L, the length of the longest straw that can fit into the can, correct to two decimal places.

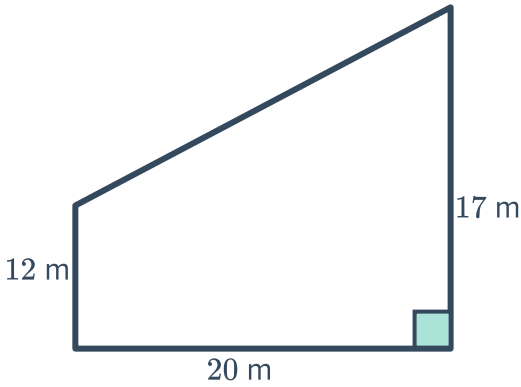


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6 Find the value of d in the following figure, correct to one decimal place:



7 Two flag posts of height $12\,\text{m}$ and $17\,\text{m}$ are erected $20\,\text{m}$ apart. Find l, the length of the string needed to join the tops of the two posts, correct to one decimal place.



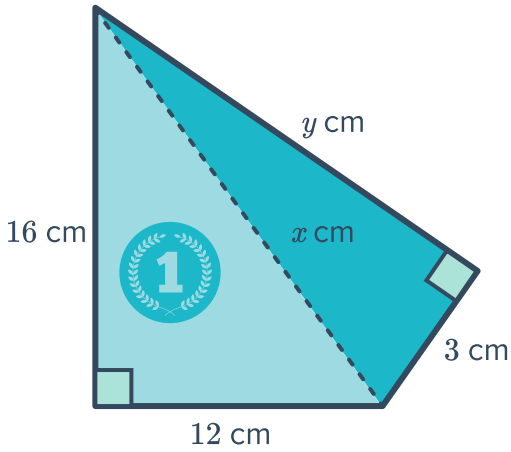
8 The town Bunderidda is $6\,\text{km}$ directly south of Appleby and $8\,\text{km}$ directly west of Cassel. Find the shortest distance from Appleby to Cassel.

9 The top of a flag pole is $4\,\text{m}$ above the ground and the shadow cast by the flag pole is $9\,\text{m}$ long. Find the distance from the top of the flag pole to the end of its shadow, correct to two decimal places.

10 A ladder of height $h\,\text{cm}$ is placed against a vertical wall. If the bottom of the ladder is $70\,\text{cm}$ from the base of the wall and the top of the ladder touches the wall at a height of $240\,\text{cm}$, find h.

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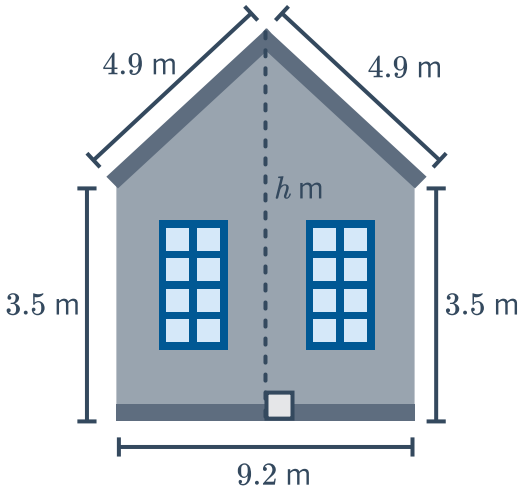
11 A sports association wants to redesign the trophy they award to the player of the season. The front view of one particular design is shown in the diagram:



a Find the value of x.

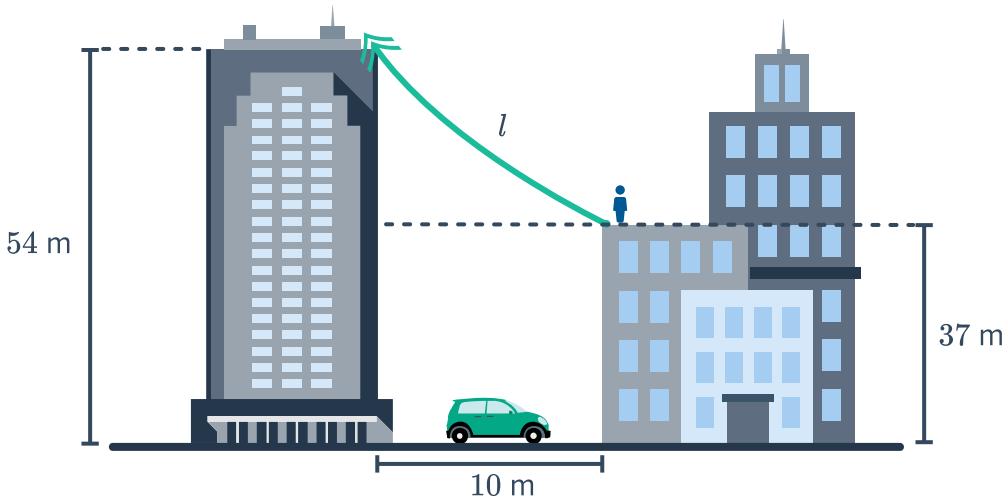
b Find the value of y, correct to two decimal places.

12 Marge's house has the outer dimensions as shown in the diagram:



Find the height of the house, h, correct to one decimal place.

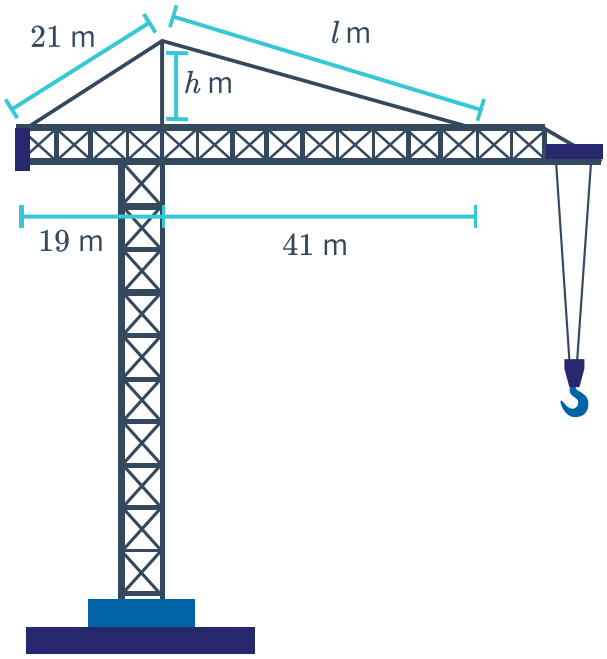
13 A movie director wants to shoot a scene where the hero of the film fires a grappling hook from the roof of one building to the roof of another. The shorter building is $37\ \text{m}$ tall, the taller building is $54\ \text{m}$ tall and the street between them is $10\ \text{m}$ wide.



Find the minimum length of rope, l, needed for the grappling hook, correct to two decimal places.

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14 Consider the crane shown below. To help bear heavier loads, a support cable joins one arm of the crane to the other, through a small tower that rises h m above the crane arm.

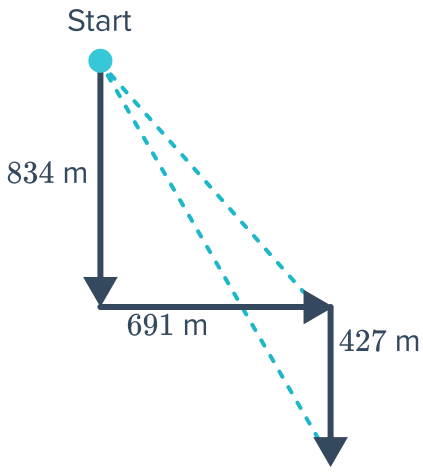


a Find the value of h, correct to two decimal places.

b Hence, find the value of l, correct to two decimal places.

c Find the length of the support cable, correct to two decimal places.

15 Emma hikes south of her starting position for 834 m and then 691 m east, before stopping for a lunch break. She then travels south again for 427 m before arriving at her final destination.

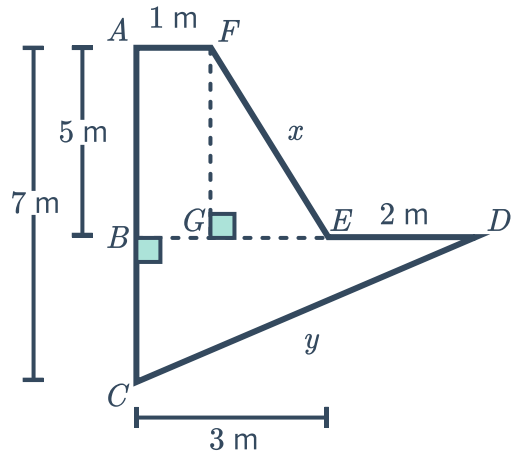


a Find the shortest distance between where Emma started and where she stopped for lunch, correct to two decimal places.

b Find the shortest distance between where Emma started and where she finished her journey, correct to two decimal places.

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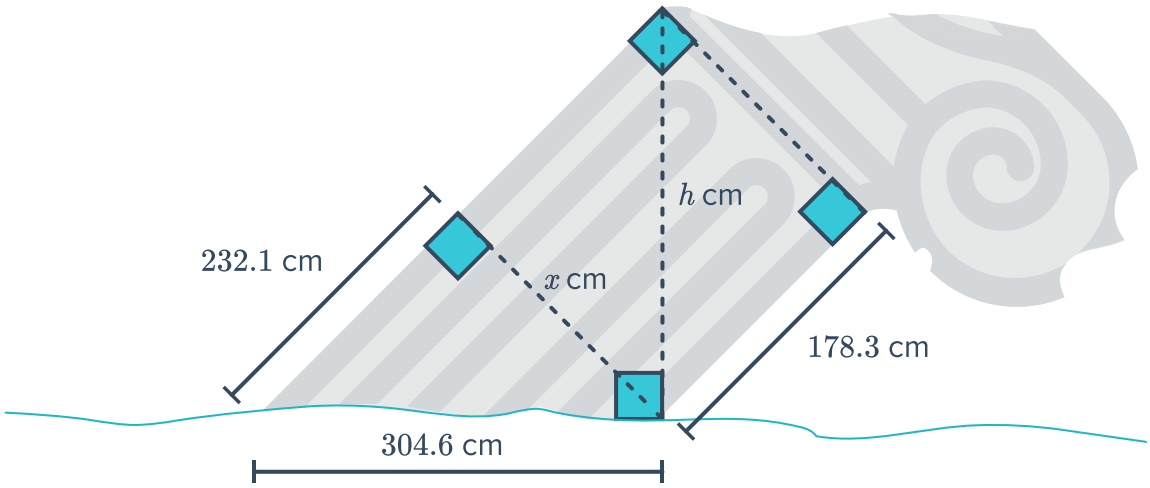
16 A farmer wants to build a fence around the entire perimeter of his land, as shown in the diagram. The fencing costs \$37 per metre.



- **a** Find the value of x, correct to two decimal places.
- **b** Find the value of y, correct to two decimal places.
- **c** How many metres of fencing does the farmer require, if fencing is sold by metre?
- **d** At \$37 per metre of fencing, how much will it cost him to build the fence along the entire perimeter of the land?

17 Archeologists have uncovered an ancient pillar which, after extensive digging, remains embedded in the ground. The lead researcher wants to record all of the dimensions of the pillar, including its height above the ground.

However, the team can only take certain measurements accurately without risking damage to the artifact. These measurements are shown in the diagram.

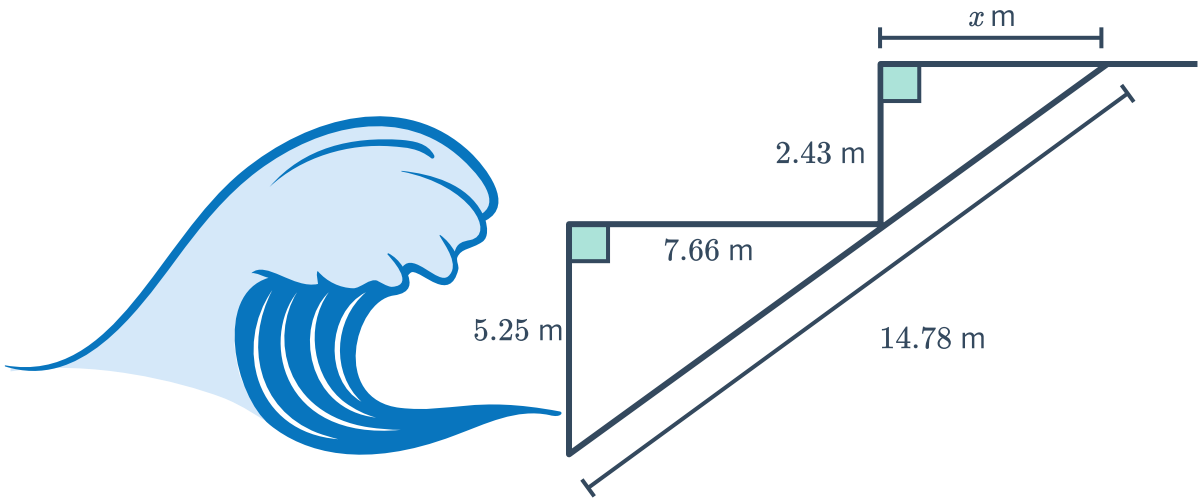


- **a** Find the value of x, correct to two decimal places.
- **b** Hence, find h, the height of the pillar, correct to two decimal places.

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18 A city council plans to build a seawall and boardwalk along a local coastline. According to safety regulations, the seawall needs to be $5.25\ \text{m}$ high and $7.66\ \text{m}$ deep and will be built at the bottom of a $14.78\ \text{m}$ long sloped section of shoreline. This means that the boardwalk will need to be built $2.43\ \text{m}$ above the seawall, so that it is level with the public area near the beach.

Find the width of the boardwalk, x, correct to two decimal places.



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